

Department of Computer Science and Engineering
Academic Year 2021– 2022(Odd Semester)
Degree, Semester & Branch: VII Semester B.E. Civil, EEE, ECE
Course Code & Title: OCS752 Introduction to C Programming
Name of the Faculty member (s): Dr.V.Anusuya, Assoc.Prof/CSE
Innovative Practice Description

- **Unit / Topic:** Unit 3 / Strings- Concatenate, Copy, Reverse, Substring.
- **Course Outcome:** The student will be able to design and implement Programs using strings
- **Topic Learning Outcome:** TLO6.
- **Activity Chosen:** One Minute Paper
- **Justification:**
 - In C string concepts such as declaration of string, initialization of string, string functions such as Concatenate, Copy, Reverse and finding substring are difficult to understand. This activity makes the students to get a comprehensive knowledge in this String concept.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The students were asked to take a paper and pen.
 - They were asked to write the concept of string and string functions.
 - The classes were in the offline mode they were asked to submit a paper.

The students also write about the specific topic which is needed to be discussed or clarified in the further classes as shown in Figure 1. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

○ **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO12	PSO1	PSO2
CO2	2	3	3	3	1	2

(1 – Low 2 – Moderate 3 – High)

○ **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO12	PSO1	PSO2
	2	3	3	3	1	2
Justification for correlation	The basic knowledge of engineering fundamentals	Basic C programming concepts using strings are used in	The student will be able to develop		The student can be able to apply C programming using string concepts in	The student can be able to develop C

	is necessary to develop C programs using strings	analysis of complex engineering problems	complex problems using C program with string concept		emerging trends in Information and Communication	programs using strings are used for developing embedded system
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• **Images / Screenshot of the practice:**

Strings
M. Harish Ram.
953618105006

1) Initialization of string array
Initializing unisized Array of character
char name[] = {'P','I','N','K','\0'}

2) Library function used for a string handling

function	work of function
strlen()	Calculate the length of string
strcpy()	Copy a string to another string
strcat()	Concatenates two string
strtol()	Convert string to lower case

3) String replacement, String insert, String index.

Strings
M. Anavindh.
953618105002

1) Initialization of string array
Initializing unisized Array of character
char name[] = {'P','I','N','K','\0'}

2) Library functions used for string handling

Function	Work of function.
strlen()	Calculate length of string.
strcpy()	Copies a string to another string.
strcat()	Concatenates two string.
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3) string replacement, string insert.
String index.

String mind
Prithvisharan
953618105005

Introduction to String
String means combination of Number of characters.
These are used to store any word in any variable of constant.

Strings

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graph TD
    Strings --> Declaration[Declaration of Strings]
    Strings --> Initialization[Initialization of Strings]
        
```

String operations:

- * str length - strlen
- * compare - strcmp
- * copy - strcpy
- * Reverse - strrev
- * String Insertion
- * String Indexing
- * Array of strings

Figure1 Sample Papers

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders**

- Students are spent in their own learning.

- Students actively participate and respect is given to every member.
- This activity encouraged the students to recollect the concept.
- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - The activity helped to know the involvement of the students in listening.
 - Students are spent in their own learning.
 - It increases class attendance and attentiveness and also improves the students' skill of writing and critical analysis of the topics.
- ***Challenges faced in implementation:***
 - Some of the students are difficult to understand string compare without using built in function.
 - The time of activity is planned is 10 minutes but it takes 15 minutes.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>
- Pradip Dey and Manas Ghosh, - Programming in C, Second Edition, Oxford University Press, 2011.

Signature of Faculty Member

HOD